Annotated slides

MA385 Part 3: Linear Algebra 1

3.3 LU-factorisation

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Tadeusz Banachiewicz (left), was polish mathematician who first presented Gaussian Elimination as matrix factorization/decomposition. Alan Turing (right) formulated LU-decomposition as we know it today and, moreover, gave the first analysis of the impact of round-off error. Photos: Wikipedia.

1. Outline of Section 3.3

In these slides,

- *LT means "lower triangular"*
- *UT means "upper triangular"*

1 A formula for LU-factorisation

2 Existence of an LU-factorisation

3 Exercises

For more, see Section 2.3 of Suli and Mayers:
https://ebookcentral.proquest.com/lib/nuig/reader.action?docID=221072&ppg=51&c=UERG

1. Outline of Section 3.3

The goal of this section is to demonstrate that the process of Gaussian Elimination applied to a matrix A is equivalent to factoring A as the product of a unit lower triangular and upper triangular matrix.

The Section 3.2 we saw that each elementary row operation in Gaussian Elimination involves replacing A with $(I + \mu_{rs}E^{(rs)})A$.

Example: For the 3×3 case, this involved computing

$$(I + \mu_{32}E^{(32)})(I + \mu_{31}E^{(31)})(I + \mu_{21}E^{(21)})A.$$

1. Outline of Section 3.3

In general we multiply A by a sequence of matrices

$$(I + \mu_{rs} E^{(rs)}),$$

all of which are **unit lower triangular** (=unit LT) matrices.
When we are finished we have reduced A to an **upper triangular** (UT) matrix.
So we can write the whole process as

$$L_k L_{k-1} L_{k-2} \dots L_2 L_1 A = U, \tag{1}$$

where each of the L_i is a unit LT matrix.

$L_k \; L_{k-1} \; L_{k-2} \; \cdots \; L_3 \; L_2 \; L_1 \; A = U$

$\tilde{L} A = U.$

1. Outline of Section 3.3

However, we know from Section 3.2 that the product of **unit LT** matrices is itself a unit LT matrix. So we can write the whole process described in (1) as

$$\tilde{L}A = U. \tag{2}$$

Also from Section 3.2, the inverse of a **unit LT** matrix exists and is a **unit LT** matrix. So we can write (2) as

$$A = LU \qquad \text{where } L = (\tilde{L})^{-1}$$

where L is unit lower triangular and U is upper triangular.
This is called "LU-**factorisation**".

1. Outline of Section 3.3

Definition 3.4.1

The LU-**factorization** of the matrix is a unit lower triangular matrix L and an upper triangular matrix U such that $LU = A$.

Example 3.4.1

If $A = \begin{pmatrix} 3 & 2 \\ -1 & 2 \end{pmatrix}$ then:

$$LU = A \Rightarrow \begin{pmatrix} 1 & 0 \\ l_{21} & 1 \end{pmatrix}\begin{pmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ -1 & 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} u_{11} & u_{12} \\ l_{21}u_{11} & l_{21}u_{12}+u_{22} \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ -1 & 2 \end{pmatrix} \Rightarrow u_{11}=3,\ u_{12}=2$$

Next $l_{21}(3) = -1 \Rightarrow l_{21} = -\frac{1}{3}$. Then $l_{21}u_{12}+u_{22}=2 \Rightarrow u_{22}=0$

Example 3.4.2

If $A = \begin{pmatrix} 3 & -1 & 1 \\ 2 & 4 & 3 \\ 0 & 2 & -4 \end{pmatrix}$ then: $LU = A$

$$\underset{L}{\begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix}} \underset{U}{\begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}} = \underset{A}{\begin{pmatrix} 3 & -1 & 1 \\ 2 & 4 & 3 \\ 0 & 2 & -4 \end{pmatrix}}$$

$$\underset{L}{\begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix}} \underset{U}{\begin{pmatrix} 3 & -1 & 1 \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}} = \underset{A}{\begin{pmatrix} 3 & -1 & 1 \\ 2 & 4 & 3 \\ 0 & 2 & -4 \end{pmatrix}}$$

Example 3.4.2

If $A = \begin{pmatrix} 3 & -1 & 1 \\ 2 & 4 & 3 \\ 0 & 2 & -4 \end{pmatrix}$ then:

$$\underset{L}{\begin{pmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 0 & l_{32} & 1 \end{pmatrix}} \underset{U}{\begin{pmatrix} 3 & -1 & 1 \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}} = \underset{A}{\begin{pmatrix} 3 & -1 & 1 \\ 2 & 4 & 3 \\ 0 & 2 & -4 \end{pmatrix}}$$

Ste

1. Outline of Section 3.3

You should find

$$\underbrace{\begin{pmatrix} 3 & -1 & 1 \\ 2 & 4 & 3 \\ 0 & 2 & -4 \end{pmatrix}}_{A} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 0 & 3/7 & 1 \end{pmatrix}}_{L} \underbrace{\begin{pmatrix} 3 & -1 & 1 \\ 0 & 14/3 & 7/3 \\ 0 & 0 & -5 \end{pmatrix}}_{U}.$$

we got this by

1. Finding row 1 of U
2. Then col 1 of L
3. Then row 2 of U
4. Thn col 2 of L
5. Finally, Row 3 of U.

2. A formula for LU-factorisation

We now want to work out formulae for L and U where

$$a_{ij} = (LU)_{ij} = \sum_{k=1}^{n} l_{ik} u_{kj} \qquad 1 \leq i, j \leq n.$$

Since L and U are triangular,

$$\text{If } i \leq j \quad \text{then} \quad a_{ij} = \sum_{k=1}^{i} l_{ik} u_{kj} \tag{3a}$$

$$\text{If } j < i \quad \text{then} \quad a_{ij} = \sum_{k=1}^{j} l_{ik} u_{kj} \tag{3b}$$

2. A formula for LU-factorisation

The first of these equations can be written as

$$a_{ij} = \sum_{k=1}^{i-1} l_{ik} u_{kj} + l_{ii} u_{ij}.$$

Notation: $\sum_{k=1}^{0} l_{ik} u_{kj} = 0$

But $l_{ii} = 1$ so:

$$u_{ij} = a_{ij} - \sum_{k=1}^{i-1} l_{ik} u_{kj} \quad \begin{cases} i = 1, \ldots, j-1, \\ j = 2, \ldots, n. \end{cases} \tag{4a}$$

And from the second:

$$l_{ij} = \frac{1}{u_{jj}} \left(a_{ij} - \sum_{k=1}^{j-1} l_{ik} u_{kj} \right) \quad \begin{cases} i = 2, \ldots, n, \\ j = 1, \ldots, i-1. \end{cases} \tag{4b}$$

Example 3.4.3

Find the LU-factorisation of

$$A = \begin{pmatrix} -1 & 0 & 1 & 2 \\ -2 & -2 & 1 & 4 \\ -3 & -4 & -2 & 4 \\ -4 & -6 & -5 & 0 \end{pmatrix}$$

2. A formula for LU-factorisation

Full details of the example: First, using (4a) with $i = 1$ we have $u_{1j} = a_{1j}$:

$$U = \begin{pmatrix} -1 & 0 & 1 & 2 \\ 0 & u_{22} & u_{23} & u_{24} \\ 0 & 0 & u_{33} & u_{34} \\ 0 & 0 & 0 & u_{44} \end{pmatrix}.$$

Then (4b) with $j = 1$ we have $l_{i1} = a_{i1}/u_{11}$:

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & l_{32} & 1 & 0 \\ 4 & l_{42} & l_{43} & 1 \end{pmatrix}.$$

Next (4a) with $i = 2$ we have $u_{2j} = a_{2j} - l_{21}u_{2j}$:

$$U = \begin{pmatrix} -1 & 0 & 1 & 2 \\ 0 & -2 & -1 & 0 \\ 0 & 0 & u_{33} & u_{34} \\ 0 & 0 & 0 & u_{44} \end{pmatrix},$$

2. A formula for LU-factorisation

then (4b) with $j = 2$ we have $l_{i2} = (a_{i2} - l_{i1}u_{12})/u_{22}$:

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 \\ 4 & 3 & l_{43} & 1 \end{pmatrix}$$

Etc....

Finished here Thursday, 5 Nov (W09.2)

3. Existence of an LU-factorisation

Not every matrix has an LU-factorisation. So we need to characterise the matrices that do.

To prove the next theorem we need the Cauchy-Binet Formula: $\det(AB) = \det(A)\det(B)$.

Eg, If L & U give the LU-decomposition of A, $(A = LU)$ then

$$\det(A) = \det(LU)$$

$$= \underbrace{\det(L)}_{1} \underbrace{\det(U)}_{\prod_{i=1}^{n} u_{ii}}$$

So, can use LU-factorization to compute $\det(A)$

3. Existence of an LU-factorisation

Theorem 3.4.1

If $n \geq 2$ and $A \in \mathbb{R}^{n \times n}$ is such that every leading principal submatrix of A is nonsingular for $1 \leq k < n$, then A has an LU-factorisation.

Proof The proof is by induction. First, let's take $n = 2$, ...

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then $A^{(1)} = (a)$

Since $(A^{(1)})^{-1}$ exists, we have $a \neq 0$.

Then $L = \begin{pmatrix} 1 & 0 \\ c/a & 1 \end{pmatrix}$ $U = \begin{pmatrix} a & b \\ 0 & d - \frac{bc}{a} \end{pmatrix}$

is the LU-factorization of A.

3. Existence of an LU-factorisation

Now assume that the theorem holds for all matrices of order up to $(n-1)$.

Let A be an n-matrix. Partition it by the last row and column.

Then, if A has an LU-factorization, it is of the form is

$$\left(\begin{array}{c|c} A^{(n-1)} & \mathbf{b} \\ \hline \mathbf{c}^T & d \end{array}\right) = \left(\begin{array}{c|c} L^{(n-1)} & \mathbf{0} \\ \hline \mathbf{w}^T & 1 \end{array}\right) \left(\begin{array}{c|c} U^{(n-1)} & \mathbf{x} \\ \hline \mathbf{0}^T & z \end{array}\right).$$

Now we need to show that all of $L^{(n-1)}$, $U^{(n-1)}$, $\mathbf{w}$, $\mathbf{x}$ and z exist.

$$LU = \left(\begin{array}{c|c} L^{(n-1)}U^{(n-1)} + 0\vec{0}^T & L^{(n-1)}x + 0z \\ \hline w^T U^{(n-1)} + (1)\vec{0}^T & w^T x + (1)z \end{array}\right)$$

Since, by induction, $A^{(n-1)}$ has an LU-factorization so $L^{(n-1)}$ & $U^{(n-1)}$ exist.

3. Existence of an LU-factorisation

Next use that $L^{(n-1)}x = b$

Since $\det(L^{(n-1)}) = 1 \neq 0$, $(L^{(n-1)})^{-1}$ exists.

Therefore $x = (L^{(n-1)})^{-1} b$ exists as well.

Then use $w^T U^{(n-1)} = c^T$.

Since $\det(U^{(n-1)}) = \det(A^{(n-1)})$ (from Cauchy-Binet). so $(U^{(n-1)})^{-1}$ exists.

Thus w^T does as well.

Then the last eqn $w^T x + z = d$

gives $z = -w^T x + d$ ✓.

Exercise 3.4.1

Many textbooks and computing systems compute the factorisation $A = LDU$ where L and U are unit lower and *unit* upper triangular matrices respectively, and D is a diagonal matrix. Show such a factorisation exists, providing that if $n \geq 2$ and $A \in \mathbb{R}^{n \times n}$, then every leading principal submatrix of A is non-singular for $1 \leq k < n$.